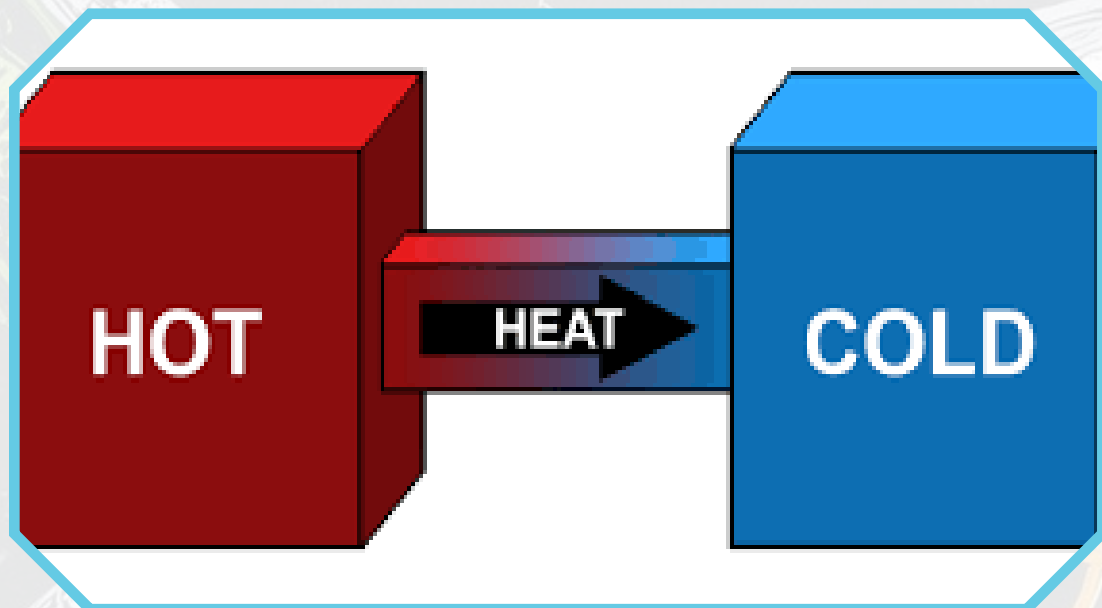




SECOND LAW OF THERMODYNAMICS



Second Law of Thermodynamics and Efficiency of Heat Engine

Introduction:

The first law of thermodynamics is a particular form of the law of conservation of energy. However, there are some limitations regarding the transformation of heat energy into mechanical energy, which are covered by the 2nd law of thermodynamics.

Second law of thermodynamics:

Lord Kelvin and Rudolf Clausius formulated the second law of thermodynamics. It consists of two statements.

Kelvin Statement:

According to this statement, "It is impossible to construct an engine, operating continuously in a cycle that can take heat from a source and convert it completely into work".

OR

In other words, it is impossible to devise an engine that would have an efficiency of 100%, even though the first law of thermodynamics will be satisfied.

Clausius Statement:

According to Clausius statement:

"It is impossible to cause heat to flow from a cold body to a hot body without the expenditure of energy".

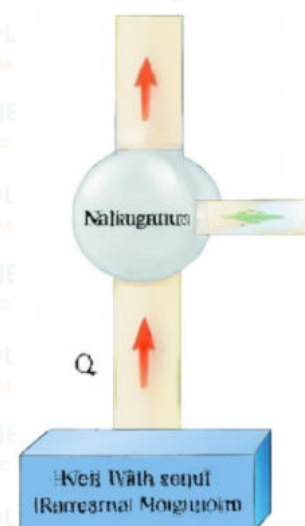
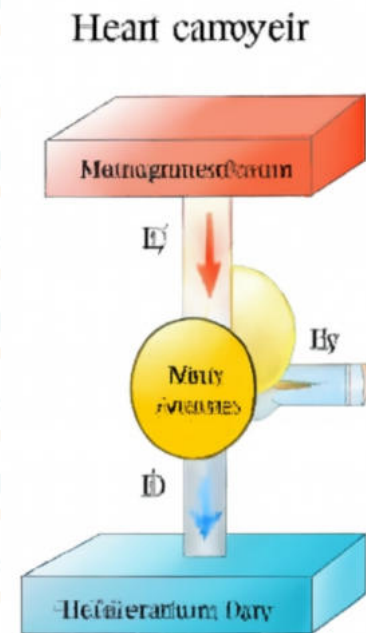
Efficiency of heat engine:

"The thermal efficiency of a cycle heat engine is defined to be the ratio of the work done by the engine in each cycle to the heat absorbed in each cycle."

$$\begin{aligned}\text{Efficiency}(\eta) &= \left(\frac{\text{output}}{\text{input}} \right) \\ \eta &= \frac{Q_1 - Q_2}{Q_1} \\ \eta &= 1 - \left(\frac{Q_2}{Q_1} \right)\end{aligned}$$

Reversible Process:

"A reversible process is one which can be retraced in exactly reverse order, without producing any change in the surroundings".



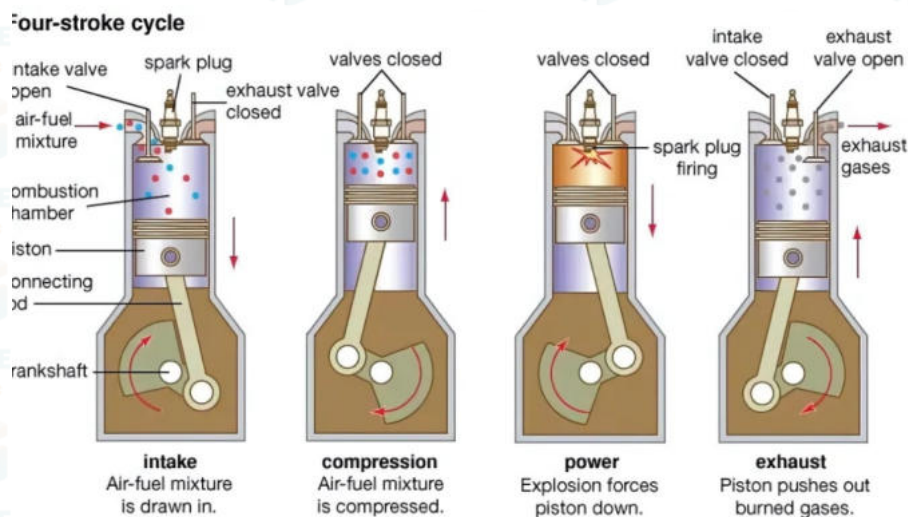
Irreversible Processes:

"The process which cannot be retraced in the backward direction by reversing the controlling factors is known as irreversible process"

Petrol Engine:

Petrol engine is an internal combustion engine designed to run on volatile fuel such as petrol (gasoline), which has spark-ignition. In these engines air and fuel are generally mixed post-compression.

It works on the Otto cycle and the name comes from the German Engineer Nikolaus Otto (1876), who made the first working prototype. Although different engines may differ in their construction technology, they are based on the principle of Carnot cycle. A typical four-stroke petrol engine also undergoes four successive processes in each cycle.

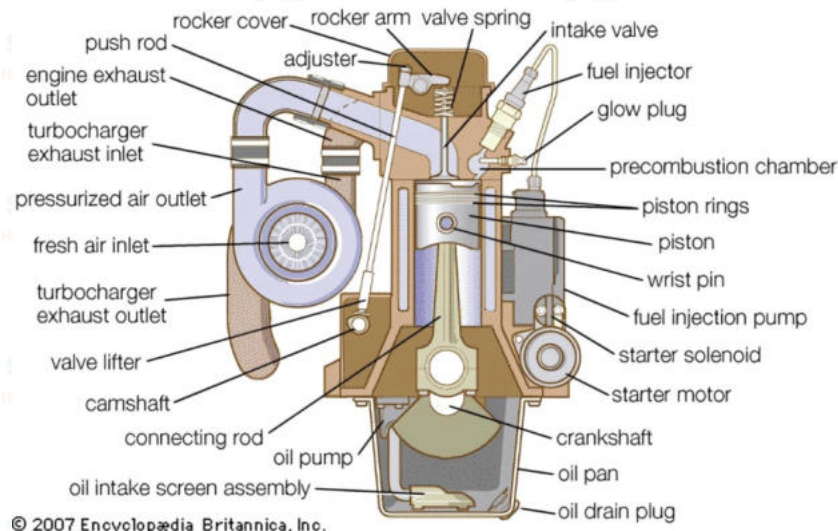


The four strokes are:

1. The cycle starts on the intake stroke in which the piston moves outward and petrol air mixture is drawn through an inlet valve into the cylinder from the carburettor at atmospheric pressure.
2. On the compression stroke, the inlet valve is closed and the mixture is compressed adiabatically by inward movement of the piston.
3. On the power stroke, a spark fires the mixture causing a rapid increase in pressure and temperature. The burning mixture expands adiabatically and forces the piston to move outward. This is the stroke which delivers power to the crankshaft to drive the flywheels.
4. On the exhaust stroke, the outlet valves open. The residual gases are expelled and piston moves inward. The cycle then begins again. Most motorbikes have one-cylinder engines but cars usually have four cylinders on the same crankshaft. The cylinders are timed to fire turn by turn in succession for a smooth running of the car. The actual efficiency of a properly tuned engine is usually not more than 25% to 30% because of friction and other heat losses.

Diesel Engine

The diesel engine named after Rudolf Diesel (German inventor and Mechanical Engineer), is an internal combustion engine. He patented his original design in 1892. When the fuel comes into contact with high temperature, it ignites, creating energy that drives the piston down transferring energy to the crankshaft. There are two classes of diesel engine: two strokes and four strokes. Most diesel engines generally use the four-stroke cycle.



No spark plug is needed in the diesel engine. Diesel is sprayed into the cylinder at maximum compression. Because air is at very high temperature immediately after compression, the fuel mixture ignites on contact with air in the cylinder and pushes the piston outward. The efficiency of diesel engine is about 35% to 40%.

Carnot Engine:

Introduction:

Carnot engine is an ideal heat engine which converts heat energy into mechanical energy. It is an imaginary heat engine free from friction & heat losses due to radiations or conduction.

Construction:

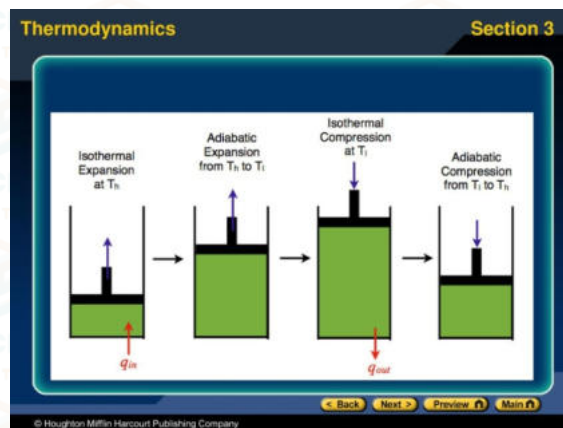
It consists of an ideal gas cylinder having conducting base, non-conducting walls and nonconducting movable frictionless piston.

Carnot Cycle:

A Carnot engine converts heat energy into mechanical energy using a cyclic process consisting of two isothermal and two adiabatic reversible processes called Carnot Cycle.

Working

Working of Carnot engine consists of steps as follows



Carnot Engine Cycle and Efficiency

Step#1: (Isothermal Expansion).

Gas in the cylinder is initially at pressure P_1 , volume V_1 , and temperature T_1 . The cylinder is then placed at a heat reservoir and absorbs heat energy (Q_1) and gas can expand isothermally, and physical variables change to P_2 , V_2 , T_1 .

Step#2: (Adiabatic Expansion).

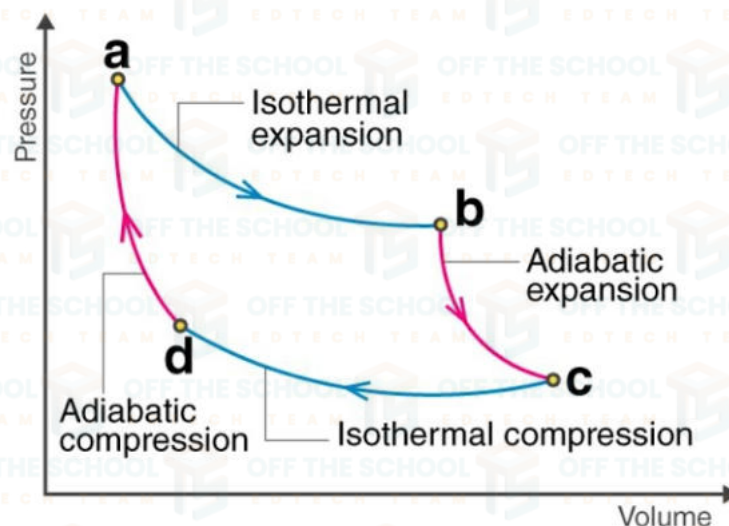
The cylinder is then placed at an insulator and external work is done, and gas can expand adiabatically, and physical variables change to P_3 , V_3 , T_2 .

Step#3: (Isothermal Compression).

The cylinder is then placed at a cold reservoir and ejects heat energy (Q_2) and gas is allowed to compress isothermally, and physical variables change to P_4 , V_4 , T_2 .

Step#4: (Adiabatic Compression).

The cylinder is then placed at an insulator and external work is done, and gas can compress adiabatically, and physical variables change to P_1 , V_1 , T_1 .



Efficiency of Carnot Engine:

Since throughout the Carnot Cycle net change in internal energy is zero, therefore, the first law of thermodynamics will be written as:

$$\Delta Q = \Delta W$$
$$\Delta Q = Q_1 - Q_2$$

Efficiency of the Carnot engine can be written as:

$$\eta = \frac{\text{Output}}{\text{Input}}$$
$$\eta = \frac{W}{Q_1}$$
$$\eta = \left(\frac{Q_1 - Q_2}{Q_1} \right)$$
$$\eta = 1 - \frac{Q_2}{Q_1}$$

Show that the efficiency of the engine increases as the ratio decreases. It can also be proved that the heat transferred to or from a Carnot engine is directly proportional to the temperature of the hot or cold body.

Thus, the efficiency of Carnot engine can be written as:

$$\eta = 1 - \frac{T_2}{T_1}$$

Refrigerator:

According to the 2nd law of thermodynamics, heat will always flow spontaneously from hot to cold body, and never the other way around.

A refrigerator causes heat to flow from cold to hot by doing work, which cools the space inside the refrigerator.

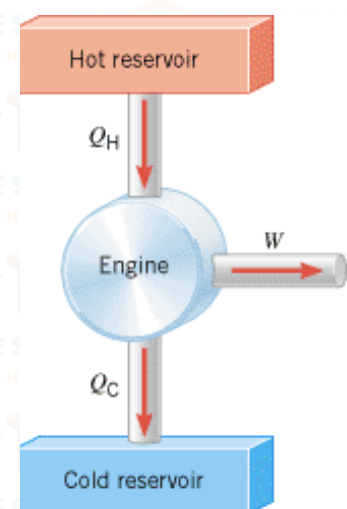
In a heat engine, the direction of energy transfer is from the hot reservoir to the cold reservoir, which is the natural direction. The role of the heat engine is to process the energy from the hot reservoir so as to do useful work.

For example, houses in summer are cooled using refrigerators called air conditioners. The air conditioner transfers energy from the cool room in the home to the warm air outside.

In Practice:

A refrigerator includes a circulating fluid that gases through two sets of metal coils that can exchange energy with the surroundings.

The fluid is cold and at low pressure when it is in the coils located in a cool environment, where it absorbs energy by heat.



The resulting warm fluid is then compressed and enters the other coils as a hot, high-pressure fluid.

There it releases its stored energy to the warm surroundings.

Efficiency of a Refrigerator and Entropy

In an air conditioner, energy is absorbed into the fluid in coils located in a building's interior; after the fluid is compressed, energy leaves the fluid through the coil. In a refrigerator, the external coils are behind or underneath the unit, while the internal coils are in the walls of the refrigerator and absorb energy from the food.

Efficiency of a Refrigerator:

The effectiveness of a refrigerator is described in terms of a number called the coefficient of performance (COP).

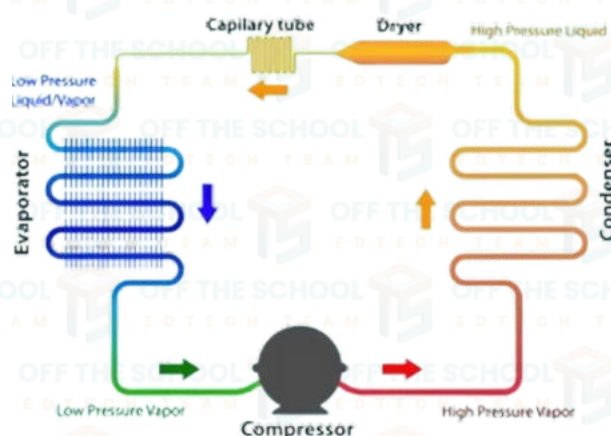
$$\text{COP of refrigerator} = \frac{\text{heat extracted}}{\text{work}}$$

$$\text{COP} = \frac{Q_c}{W}$$

$$Q_h = Q_c + W$$

$$W = Q_h - Q_c$$

$$\text{COP} = \frac{Q_c}{(Q_h - Q_c)}$$



We know that for Carnot cycle $Q_1 \propto T_1$ and $Q_2 \propto T_2$

$$\text{COP} = \frac{T_c}{(T_h - T_c)}$$

Where:

T_c is the cryogenic temperature at which the heat is removed.

T_h is the temperature at which the heat is rejected.

It can also be seen from above relations that the coefficient of performance of a refrigerator can be larger than 100%, unlike the efficiency of a heat engine which is always less than 100%. A good refrigerator should have a high COP, typically 5 or 6.

Entropy:

"The measure of a system's thermal energy per unit temperature that is unavailable for doing useful work."

$$\Delta S = \frac{\Delta Q}{T}$$

The SI unit of entropy is JK⁻¹. If heat is removed from a system, the change in entropy is written as negative.

Increase in Entropy Means Degradation of Energy:

Suppose a quantity of heat Q in a reservoir at temperature T_1 . Let the temperature of the coldest available reservoir be T_0 . A Carnot engine working between the temperatures T_1 and T_0 can absorb heat Q at temperature T_1 and do useful work W_1 given by:

$$W_1 = Q - Q_2$$

Carnot Engine Efficiency and Energy Degradation

$$\eta = \frac{W_1}{Q} = 1 - \frac{T_0}{T_1}$$

$$W_1 = Q \left(1 - \frac{T_0}{T_1} \right) \text{ ----- 1}$$

Equation 1 gives maximum available energy which can be converted into useful work when heat Q is stored in the reservoir at temperature T_1 .

Consider an irreversible process, in which heat Q flows from the reservoir at temperature T_1 to another reservoir at a lower temperature T_2 . A Carnot engine working between the temperatures T_2 and T_0 can now take heat Q from the reservoir at temperature T_2 , and do useful work W_2 given by:

$$\eta = \frac{W_2}{Q} = 1 - \frac{T_0}{T_2}$$

$$W_2 = Q \left(1 - \frac{T_0}{T_2} \right) \text{ ----- 2}$$

Equation 2 gives the maximum available energy which can be converted into useful work. When heat Q is stored in the reservoir at lower temperature T_2 . As $T_2 < T_1$, we see from equation 1 and 2 that W_2 is less than W_1 , i.e. available energy decreases with the increase of entropy during an irreversible process.

Since all natural processes are irreversible, we conclude that the energy of the universe is continuously becoming unavailable for useful work. This is called the degradation of energy.

From the equation 1 and 2, we get:

$$W_1 - W_2 = Q \left(\frac{T_0}{T_2} - \frac{T_0}{T_1} \right)$$
$$W_1 - W_2 = T_0 \left(\frac{Q}{T_2} - \frac{Q}{T_1} \right)$$

Now $\frac{Q}{T_1}$ decreases in entropy of the reservoir at temperature T_1 and $\frac{Q}{T_2}$ increases in entropy of the reservoir at temperature T_2 .

Thus $\left(\frac{Q}{T_1}\right) / \left(\frac{Q}{T_2}\right)$ is increase in entropy of the universe. Also $(W_1 - W_2)$ is the amount of energy which has been degraded or made available for useful work.

Hence, the equation shows that the increase of unavailable energy is equal to the increase in entropy of the universe multiplied by the temperature of the coldest available reservoir.

Using equation, as entropy increases, so unavailable energy also increases, and so the useful energy decreases. It is called degradation of energy.

Energy is Degraded During All-Natural Processes:

It is believed that the temperature of the universe is increasing gradually and its entropy is also increasing. The mode of increase of entropy indicates that after a very long time, the universe will reach a state of maximum entropy.

Systems Tend to Become Less Orderly Over Time

Entropy of the universe will be maximum and all the objects will be at the same temperature. At that time, no useful energy will be available and life will cease to exist, which is known as the "Heat Death" of the universe, which may not occur in the near future.

Systems Tend to Become Less Orderly Over Time:

The disorder of the system increases during any natural process, a stage will reach when the universe will approach a stage of maximum disorder. At this stage, matter will become a uniform mixture, i.e., the whole universe will be at uniform temperature. No work can then be done and heat flow will cease.

In all the processes going on in practical life, the entropy always increases with time, and disorder is produced more and more. Hence, this increase of entropy is an identification for the time passage.